

ARCHITECTURE REPORT

1. REQUIREMENTS

```
{
  "system_name": "SimpleInventoryService",
  "architecture_style": "Layered Architecture",
  "functional_requirements": [
    "Add products",
    "Update product quantity",
    "View product list"
  ],
  "non_functional_requirements": {
    "scalability": "Low",
    "availability": "Medium",
    "latency": "Low",
    "reliability": "Medium"
  }
}
```

2. ARCHITECTURE (ACME ADL)

```
System SimpleInventoryService {
  Property view : String = "runtime";
  Component ProductService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Handles product related operations";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ProductRepository = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Stores and retrieves product data";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ProductController = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Handles HTTP requests and calls Product Service";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component InventoryServiceAPI = new Component {
    Port in;
    Port out;
```

```
Property responsibility : String = "Exposes API endpoints for client interactions";
Property state : String = "stateless";
Property scaling : String = "horizontal";
}
Attachment ProductController.out to ProductService.in {
Property delivery : String = "at-least-once";
Property ordering : Boolean = true;
Property backpressure : String = "bounded";
}
Attachment ProductService.out to ProductRepository.in {
Property delivery : String = "at-least-once";
Property ordering : Boolean = true;
Property backpressure : String = "bounded";
}
Attachment ProductController.out to InventoryServiceAPI.in {
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry + CircuitBreaker";
Property failure_isolation : Boolean = true;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
Property encryption : String = "in-transit";
Property constraint_scalability : String = "Low";
Property constraint_availability : String = "Medium";
Property constraint_latency : String = "Low";
Property constraint_reliability : String = "Medium";
}
```